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Abstract

Health is a top priority, especially given that cardiovascular diseases (CVDs) remain the leading cause of death in the Philippines, affecting one in six Filipinos and accounting for 20% of all deaths. To combat this, healthcare professionals are implementing community-based programs in rural areas, though patient profiling is still manual. "Pitik," a Cebuano-Binisaya intent-based chatbot, was developed to gather information, assess cardiovascular risks, and profile individuals. Guided by Gricean Maxims, Pitik detects communication violations, ensuring effective interaction. This study used Action Research, fostering collaboration between researchers, experts, and end-users. Through three software development iterations, the Diag-Ex framework with Pre-Intent and Post-Intent Matching algorithms significantly improved Pitik's ability to respond smoothly to user prompts.

Keywords

cardiology, cardiovascular, chatbot, communication, networking and broadcast technologies, computing and processing, conversational ai

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Abstract

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Introduction

Cardiovascular diseases (CVDs) remain a significant public health challenge in the Philippines, consistently ranking as the leading cause of mortality (Cacciata et al., 2021). These diseases account for approximately 20% of all deaths and 35% of premature deaths in the country, affecting one in every six Filipinos. High blood pressure, high cholesterol, and smoking are key risk factors, along with other conditions and lifestyle choices such as diabetes, obesity, and overweight. To address these issues outside Metro Manila, one of the initiatives of healthcare professionals is conducting community outreach programs in rural and underserved areas. These initiatives involve profiling residents and providing essential healthcare advice, aiming to mitigate the impact of cardiovascular-related diseases in these communities (Reyes et al., 2023). Community-based profiling and diagnosing of patients involves localized digital systems and at times physical records depending on the institution. Health records between institutions are encoded physically through paper and are not standardized between institutions (Evans, 2016; Menachemi, 2011). In many cases, health records are maintained using localized digital systems, often with heavy reliance on physical records, especially in less developed areas. The inconsistency in how patient information is recorded and stored across different institutions leads to significant discrepancies, making it difficult to transfer patient data effectively. This fragmentation of data hinders the ability to compile a comprehensive medical history, which is crucial for accurate diagnosis and treatment. While some healthcare providers may have advanced electronic health records (EHR) systems, others may still depend on outdated or manual methods. This disparity not only complicates data sharing but also limits the overall efficiency and accuracy of patient profiling (Quinn et al, 2019; Casey et al., 2016; Iyanna et al, 2022).

The Philippines offers various healthcare applications like KonsultaMD, which provides 24/7 access to licensed doctors via telehealth (Noceda et al, 2023). While these technologies expand healthcare access, they lack crucial elements. None are exclusively available in Cebuano-Binisaya, a language spoken by millions,

particularly in rural areas where access to healthcare is already limited. This gap is significant, as language barriers can hinder effective communication and understanding, especially in areas with high rates of cardiovascular diseases. Moreover, there are no existing Cebuano-Binisaya healthcare datasets, making it challenging to create targeted digital tools for these communities. Developing a Cebuano-Binisaya chatbot is essential to bridge these gaps, providing culturally and linguistically appropriate healthcare support, and ensuring that vital health information and services are accessible to those who need them most.

To bridge these gaps and bring critical healthcare support to those who need it most, we developed Pitik—a web application designed to speak directly to the heart of the Cebuano-Binisaya community. Pitik is more than just a chatbot; it is a lifeline that helps users navigate their cardiovascular health concerns in their native language, ensuring that no one is left behind. By focusing on underserved populations in remote areas, Pitik aims to bring the benefits of modern healthcare technology to those who have been overlooked for too long.

There are several applications with functionalities like Pitik, each offering unique approaches. For instance, Diabot is a predictive medical chatbot that provides users with general disease and diabetes predictions, encouraging proactive measures such as lifestyle changes and medication adjustments (Sarma et al., 2019). Diabot employs ensemble learning to deliver accurate diagnoses, utilizing datasets such as a general health dataset for disease prediction and the Pima Indian diabetes dataset for diabetes prediction. Other chatbots collect personal and medical details, which are then analyzed using algorithms like Support Vector Machine to identify specific illnesses and suggest treatment options. Additionally, some chatbots follow a linear design, progressing from symptom extraction to mapping and diagnosis, with severe conditions triggering a referral to a doctor, who is then provided with relevant patient details from the database.

The availability of data is crucial for developing technology-based solutions, particularly those that heavily rely on data for their functionality. Moreover, the Cebuano-Binisaya language lacks digitized datasets related to heart disease diagnosis or patient profiling. This absence may be attributed to the challenges in translating medical terminology into Cebuano-Binisaya and the potential under-exploration or unavailability of the language’s implementation in the medical field, particularly in digital formats accessible to the public. Also, the existing data in this domain is predominantly standardized forms, which necessitates constructing a new dataset from the ground up. This process introduces various challenges, including sourcing the data, determining its scope, and ensuring thorough data cleaning and refinement. Equally important in developing a chatbot is its internal structure. How responses are formulated can significantly influence users’ perceptions, their interpretation of the information, and their subsequent actions. Research indicates that effective advice depends not only on the content but also on the presentation of that content. This highlights the importance of evaluating chatbot responses in terms of both content and structure to ensure successful and meaningful communication.

Considering these challenges, the primary objective of this study is to develop Pitik, a chatbot designed for heart disease-related inquiries in the Cebuano-Binisaya language. The project aims to achieve this by building a schema-guided dialogue dataset specifically for heart disease-related illnesses in Cebuano-Binisaya, addressing issues related to the internal structure of the chatbot, and discovering a conversational scheme that aligns with Cebuano-Binisaya’s conversational flow. Furthermore, the study seeks to employ both intrinsic and extrinsic evaluation methods to assess Pitik’s performance based on the newly developed conversational scheme, ensuring that the chatbot effectively communicates and meets user expectations.

Methods

The development of Pitik adhered to an Iterative Software Development process, allowing for continuous enhancements and refinements. As illustrated in Figure 1, this process facilitated systematic progress through multiple stages. In the first iteration, the team collaborated with healthcare professionals and reviewed existing literature to gather comprehensive requirements. This informed the design of a user-friendly interaction medium in Cebuano, ensuring cultural and linguistic relevance. The initial implementation of the chatbot was achieved using Google’s DialogFlow, based on a risk assessment form provided by healthcare

experts. Early testing underscored the need for additional data and further expert input, highlighting areas for significant improvement. These insights drove subsequent iterations, allowing Pitik to evolve into a more robust and effective tool for cardiovascular risk assessment and patient profiling.

In the second iteration, the design was meticulously refined to align more closely with Gricean Maxims, aiming to minimize errors and reduce user confusion. To further enhance the conversational structure, suggestion chips were introduced, providing users with clear response options and streamlining interactions. The testing phase incorporated valuable feedback from both users and linguistic experts, allowing for a comprehensive review of the chatbot's conversation flow. This rigorous evaluation led to targeted improvements, ensuring a more intuitive and effective user experience.

The third iteration concentrated on optimizing the conversational flow to better accommodate the nuances of the Cebuano-Binisaya language, addressing challenges identified in earlier stages. This phase involved refining the chatbot's ability to handle a broader spectrum of user inputs, enhancing its flexibility and responsiveness. The evaluation process was rigorous, employing the Analytic Hierarchy Process (AHP) method to assess overall quality. Pitik underwent both intrinsic and extrinsic evaluations: intrinsically, its thoroughness in assessing cardiovascular health was scrutinized, while extrinsically, it was evaluated based on user feedback from prior iterations. These assessments ensured that Pitik not only met the technical requirements but also resonated with its target users, enhancing its effectiveness as a healthcare tool.

The researchers made use of Grice's Maxims of conversation to evaluate the acceptability of the user input which plays a very important role in the success or failure of Pitik. Grice's maxims of conversation were chosen as a framework since it serves as a guide in constructing and evaluating the design of our conversational flow. The philosopher Paul Grice (1975) proposed four conversational maxims which serve as a way to explain the link between utterances and what is understood from them. It is based on his cooperative principle which is pragmatic in its approach and is so-called because listeners and speakers must speak cooperatively and mutually accept one another to be understood in a particular way. There are four main maxims proposed by linguist Paul Grice to describe principles that people intuitively follow to guide their conversations. This work is guided by this principle to have effective communication with the users as shown in Figure 2. After the testing phase, the user's answers were evaluated using this study to see if the chatbot and user have established successful communication. Violations to the Gricean Maxim involve not following the designated rules shown below, violations may overlap where a sentence or phrase can violate multiple maxims.

Gricean Maxims:

1. The maxim of quantity - where one tries to be as informative as one possibly can, and gives as much information as is needed, and no more.
2. The maxim of quality - where one tries to be truthful and does not give information that is false or that is not supported by evidence
3. The maxim of relation - where one tries to be relevant and says things that are pertinent to the discussion.
4. The maxim of manners - when one tries to be as clear, as brief, and as orderly as one can in what one says, and where one avoids obscurity and ambiguity.

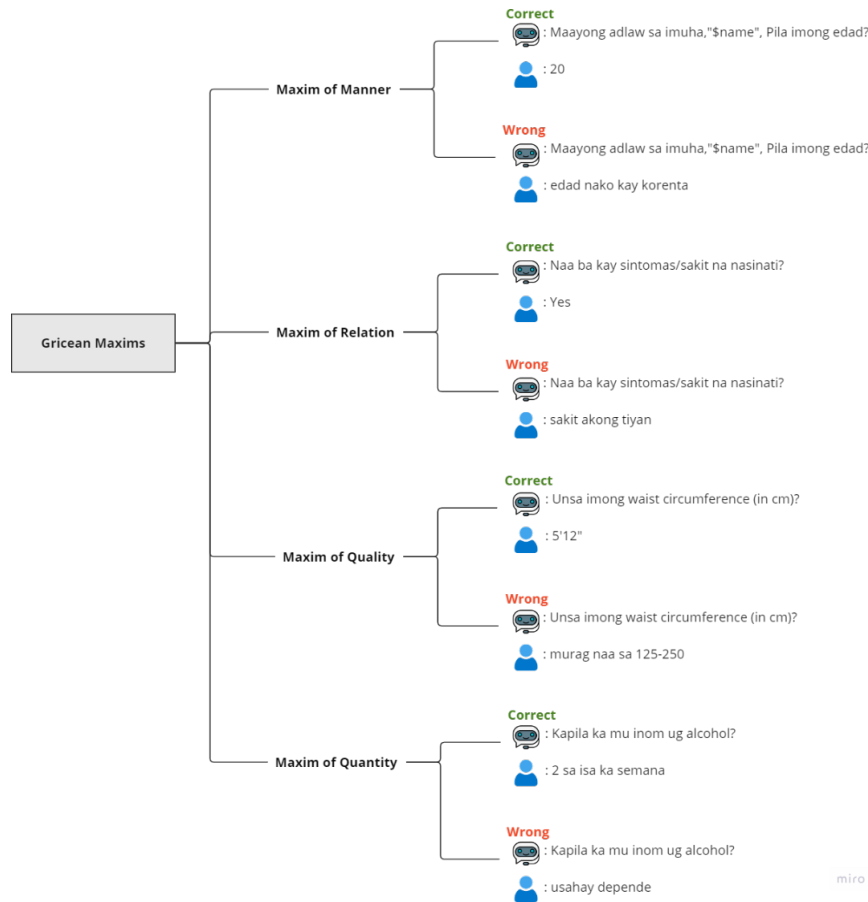


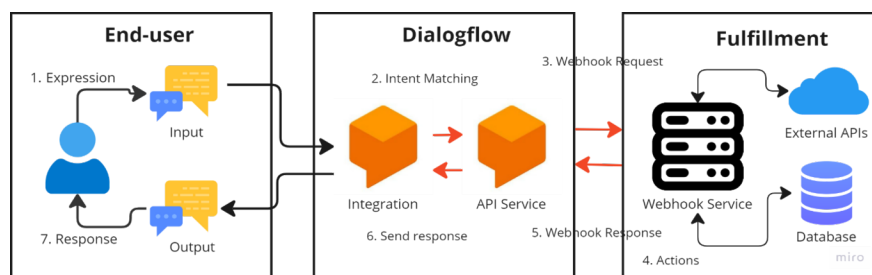
Figure 2: Example of Grice Maxim Occurrences in Pitik

As Pitik underwent iterative testing, it was progressively trained to comprehend the optimal and most tolerable responses gathered from users during the Data Collection phase. Currently, Pitik operates in alignment with the principles of the Cooperative Principle, particularly Grice's Maxims, while also drawing on the insights from our collected data. Violations of Grice's Maxims are triggered whenever DialogFlow fails to match a user input with a defined intent, resulting in a fallback intent. This phenomenon predominantly affects Pitik's generic questions regarding exercise, diet, smoking, and alcohol, which often elicit a wide range of user responses, frequently resulting in the flouting of the Cooperative Principle. As DialogFlow matches the closest training phrases, it struggles to address variations in user responses. To enhance functionality, developers utilize intents to define specific tasks that users can perform, ensuring a more structured interaction.

Process Flow of Pitik using DialogFlow

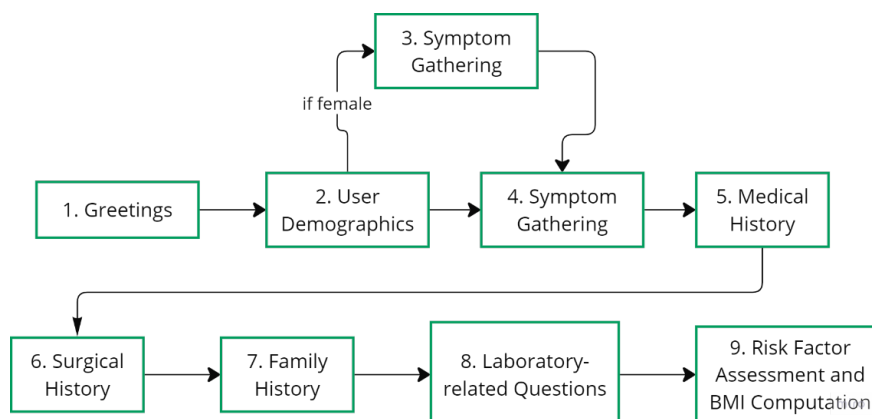
The researchers utilized DialogFlow, a natural language understanding platform, to design and integrate a conversational user interface for the Pitik chatbot. The platform facilitates user interaction by allowing end-users to input text, which DialogFlow matches to specific intents while extracting relevant parameters. Once the intent is identified, DialogFlow sends a webhook request to the designated service, including information about the matched intent, the associated action, parameters, and the predefined response. Pitik then executes the necessary actions, retaining the extracted information and guiding users through the relevant conversational flow. Subsequently, Pitik generates a webhook response message directed back to DialogFlow, containing the response intended for the end-user. Finally, DialogFlow relays this response to

the user, ensuring seamless interaction and effective communication throughout the conversation.



Cardiovascular Risk Assessment Sheet

Pitik initiates the conversation by greeting the user, establishing a friendly and welcoming environment. Following the exchange of greetings, Pitik collects essential demographic information, including the user's name, age, sex, weight, and height. For female users, the chatbot then inquires about pregnancy-related details, such as current pregnancy status and the number of children they have. After gathering this information, Pitik proceeds to collect information about the user's symptoms, as well as their medical, surgical, and family histories. Subsequently, the chatbot poses questions related to laboratory results, including measurements such as HbA1c, systolic blood pressure, and diastolic blood pressure. During the contact details stage, users have the opportunity to provide their email addresses for receiving comments or recommendations from healthcare professionals. Finally, Pitik generates a risk factor assessment and calculates the Body Mass Index (BMI) for the user. The complete conversational flow of Pitik is illustrated in Figure 4.



To assess the cardiovascular risk of Pitik user, the researchers utilized a cardiovascular risk assessment sheet [11https://pcna.net/wp-content/uploads/2018/12/1-patient_assessment.pdf](https://pcna.net/wp-content/uploads/2018/12/1-patient_assessment.pdf)

recommended by a healthcare professional conducting community-based profiling and risk factor assessment. Risk factor is then computed using the Framingham formula (D'Agostino et al., 2008):

$$\text{Risk Factors} = (\ln(\text{Age}) * 3.06117) + (\ln(\text{Total.cholesterol}) * 1.12370) - (\ln(\text{HDL.cholesterol}) * 0.93263) + (\ln(\text{Systolic.blood.pressure}) * \text{On.blood.pressure.medication}) + \text{Cigarette.smoker} + \text{Diabetes.present} - 23.9802$$

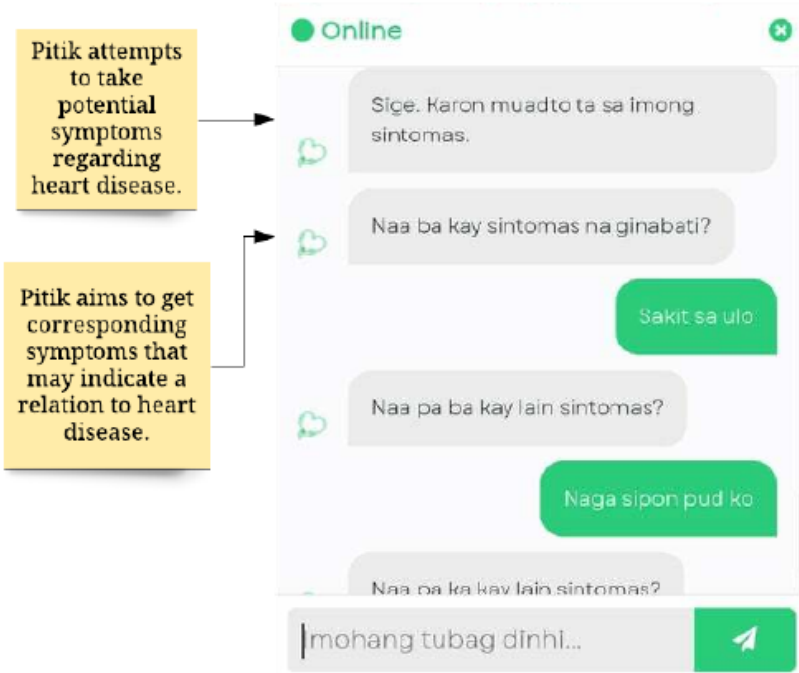
$$\text{Risk} = 100 * (1 - 0.88936^{e(\text{Risk_Factors})})$$

Pitik is trained to recognize input symptoms based on a predefined list of cardiovascular related diseases and its corresponding symptoms ¹¹<https://my.clevelandclinic.org/health/diseases/21493-cardiovascular-disease>. Table 1 shows the cardiovascular diseases and their corresponding symptoms (in Cebuano-Binisaya).

Table 1: Cardiovascular Diseases and Symptoms

Cardiovascular Disease	Symptoms
Congenital Heart Disease	Pagkabluhe, namutla, hubag, hupong,lisud ginhawa, wad-an kusog, pitik, di regul
Arrhythmia	Gaan - Paminaw, Bloat - Tiyan, Paspas - Heartbeat, Hinay - Pulso, Taas - Blo
Cardiomyopathy	kapoy, lisud og ginhawa, hubag ang bitiis, bukong ug tiil, bloat nga tiyan, ubo, s
Atherosclerosis	kabugnaw,manhid,Kahuyang,kawad-an og usog
Coronary Heart Disease	Sakit sa dughan, Dili komportable ang dughan, Kumot sa dughan, Kakulang sa
Heart Infection	Kakapoy, Kalipong, Pagkulbakulba Sakit sa dughan, Kalintura, Nagamaga akon

Figure 5 shows an actual example of how Pitik attempts to capture user symptoms.



Analytic Hierarchy Process

Pitik’s testing phase centered on evaluating the chatbot’s overall quality using the method recommended by (Radziwill and Benton, 2017). This approach employs the Analytic Hierarchy Process (AHP), a structured technique for organizing and analyzing complex decisions that involve both qualitative and quantitative factors. The researchers selected AHP to assess and quantify criteria related to the quality of the Pitik chatbot effectively. The goal was to compare and evaluate the performance of the original version of Pitik against the updated version per iteration. The criteria chosen for this assessment were aligned with the study’s thematic focus and objectives shown in Figure 6.

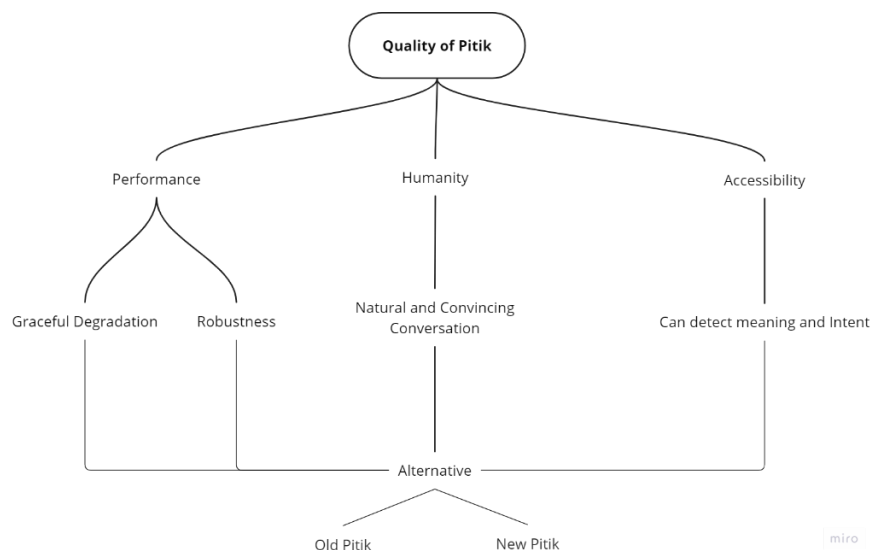


Figure 6: Hierarchical Structure for Pitik Chatbot Evaluation

Each category has its different methods of evaluation to contribute to the overall chatbot quality. For the category ‘Humanness’, the evaluation made use of a google form comprising an answerable, open-ended question form and a Likert-scale format question from appearing at the very end of the chat conversation that lets users be able to evaluate the chatbot based on the category that is identified to analyze the quality producing Likert scale information and opinionated information about the user-opinion. The researchers utilized a tally system for counting Gricean Maxim Violations for the user inputs and a linguist expert-reviewed verdict. For the ‘Performance’, the researchers used another tally system that counts the number of times Pitik has failed to respond appropriately during the conversation with the user. For the ‘Accessibility’, the researchers used a similar method with finding the Performance with the difference of tallying the circumstances when the users are the ones to ask the questions for Pitik.

We have used Precision, Recall, F1 Score, Accuracy and its weighted average to evaluate our post-intent matching algorithm which trained several machine learning models to classify intents. Precision, being the ratio between the True Positives and all the Positives, enables us to identify true positives with respect to the overall positive predictions. A high precision means it has a lower false positive rate. Recall is the ratio of the correctly predicted positive occurrences with respect to the overall observations in the actual class. F1 Score is the average from both Precision and Recall and in return takes account of false positives and false negatives. Accuracy is the number of correct predictions with respect to the overall data. This is achieved by adding both true positives and true negatives to true positives, true negatives, false negatives and false positives. The weighted average is generated by multiplying each of the evaluation measurements by their corresponding weight which is the percentage count of the total class with respect to the overall test data.

Results and Discussion

The researchers were able to build a schema-guided dialogue set gathered from 100 participants by reaching through social media and word of mouth. The selection of participants preferred to be of those residing in provincial areas but due to limited time and accessibility, the selection was made if the participant could communicate using the identified language of Pitik. The first 50 participants interacted with Pitik during the second iteration then the remaining 50 participants during the third iteration. A link was sent with instructions and information concerning the purpose of the study to the participants.

The researchers gathered the responses and were able to group and analyze the data by checking which responses violated Gricean Maxims with the help of an expert. An example is shown in Table 2, which

produced bugs in the system and how the participants were able to answer specifically the laboratory-related questions such as blood pressure.

Table 2: Gricean Maxims Violations

Gricean Maxims	Sample Responses	2 nd Iteration
Manner	“murag naa sa 125-250”, “sauna naa pero karon wala”	44
Relation	“wala kay ginabutang ra nko sa kamot”, “gipaak kog ilaga, dugay na sukad bata pa ko”	18
Quantity	“oo pag ma stress ko”, “dli kay weak”	24
Quality	“murag naa sa 125-250”, “5’2 ata”	7

Several models were trained to develop an effective classifier for intent recognition. The researchers evaluated and compared the performance of four algorithms: Naive Bayes (NB), Support Vector Machine (SVM), Multi-Layered Perceptron (MLP), and Recurrent Neural Network (RNN). These algorithms are well-established for classifying text data. Naive Bayes classifiers operate under the assumption of strong (or naive) independence between the attributes of data points. They are widely used in applications such as spam filtering, text analysis, and medical diagnosis. Support Vector Machines (SVM) are supervised machine learning models designed for binary classification tasks, effectively separating data into two distinct groups. The Multi-Layered Perceptron (MLP) classifier relies on an underlying neural network to perform classification, allowing it to learn complex patterns in the data. Recurrent Neural Networks (RNNs), on the other hand, are commonly employed in speech recognition and natural language processing due to their ability to recognize sequential patterns in data and predict future scenarios, although they are typically more complex to implement.

Table 3 presents a comparative analysis of the four classifiers across two domains: Diet/Exercise and Smoking/Alcohol. All models were tested on a balanced dataset comprising 40 records derived from a total of 200 training samples.

Table 3: Model Performances for sample diet/exercise and smoking/alcohol areas

Area of Comparison	Models	Precision	Recall	F1 Score	Accuracy
Diet/Exercise	NB	71	70	70	70
	SVM	70	70	70	70
	MLP	65	65	65	65
	RNN	58	57	57	57
Smoking/Alcohol	NB	72	68	66	68
	SVM	78	72	71	73
	MLP	70	68	66	68
	RNN	61	60	59	60

The Naive Bayes and SVM models demonstrate similar performance in the Diet/Exercise category, with each achieving Precision, Recall, F1 Score, and Accuracy values of approximately 70%. This suggests that these models are relatively effective in classifying or predicting outcomes related to Diet and Exercise. The MLP model shows a slight decline in performance, with all metrics standing at 65%, indicating a moderately lower ability to generalize in this context. Conversely, the RNN model exhibits the weakest performance, with metrics in the range of 57-58%, highlighting significant challenges in capturing the patterns associated with Diet and Exercise. In the Smoking/Alcohol category, the SVM model emerges as the most effective, boasting the highest Accuracy (73%) and relatively high values across all other metrics, which underscores its robustness in handling this classification task. The Naive Bayes model, while strong in Precision (72%), shows a lower F1 Score (66%), indicating some trade-offs between Precision and Recall. MLP’s performance

mirrors that of Naive Bayes in this context, while the RNN model continues to underperform, with all metrics falling around 59-61%. The results indicate that the SVM model is particularly well-suited for tasks related to Smoking/Alcohol, outperforming other models by a notable margin. Its ability to maintain high Accuracy and balanced metrics suggests that it can be relied upon for more accurate predictions and classifications in this domain. On the other hand, the RNN model’s consistent underperformance across both areas suggests that it may struggle with the type of data or features used in these tasks.

The researchers assessed accessibility by examining the success rate of five critical laboratory questions, such as HBA1C and Systolic Blood Pressure, which often pose significant barriers due to the need for prior medical knowledge and specific instrument results. Initially, the average success rate was a mere 22%, as most respondents lacked their lab results, leading to widespread failure in answering these questions. However, in the third iteration, the introduction of the Naive Bayes algorithm and post-intent processing significantly boosted both performance and accessibility, with rates soaring from 65% to 79% and 22% to an impressive 96%, respectively. This remarkable improvement was driven by the third iteration’s enhanced user guidance and response processing, featuring new tools like suggestion chips and tooltips. The researchers recognized that many users sought definitions for medical terminologies, such as HBA1C, prompting the integration of tooltips to assist users unfamiliar with terms like Systolic Blood Pressure, Diastolic Blood Pressure, Total Cholesterol, and HDL Cholesterol. Additionally, suggestion chips were implemented to address instances where the chatbot failed to recognize user input, guiding users towards a format the system could understand. For example, when a user responded “n/a” to a question about waist circumference, a suggestion chip appeared to offer guidance. These enhancements were crucial in overcoming the limitations encountered during the training phase, ensuring that the system could adapt to unforeseen inputs and vastly improving overall user experience.

There are two key scenarios where suggestion chips are strategically deployed to enhance user interaction. In the first scenario, suggestion chips appear during the initial occurrence of a question to confirm the accuracy of user input. These questions typically involve a prompt to verify that the chatbot has correctly understood the input. In the second scenario, suggestion chips address questions that may elicit unexpected or unanticipated responses, which became evident during the system’s second iteration. For instance, when Pitik asked for the father’s name, one user out of 100 responded that they did not know their father. To accommodate such rare but important cases, the researchers created an Intent that acknowledges and skips the question. However, due to the limited occurrence of such responses, there weren’t enough training phrases to fully support this Intent. Therefore, suggestion chips were implemented at the first occurrence of such questions, enabling users to skip questions related to parents they do not know. This adjustment significantly improved performance by allowing the chatbot to handle a broader range of responses while maintaining focus on essential information.

Moreover, when users encountered one of the five laboratory questions—HBA1C, Systolic and Diastolic Blood Pressure, Total Cholesterol, and HDL Cholesterol—many did not know their results. Instead of persisting with these questions, the researchers provided suggestion chips, allowing users to select which laboratory questions they could answer. These enhancements not only improved user experience but also led to a significant increase in accessibility, as the chatbot became more adept at catering to diverse user needs and circumstances.

The data collected from the respondents were evaluated using AHP to check for version improvements. Performance values were taken from responses of Pitik that went into a fallback intent and did not continue towards further intents. Humanness was assessed from their responses and how they rated Pitik according to their experience. Accessibility was assessed as well in the same manner as to how the Performance was checked with the focus of user responses pertaining to accessibility.

Table 4: Reciprocal matrix for pairwise comparisons

Category	Performance	Humanness	Accessibility
Performance	1	9	7

Category	Performance	Humanness	Accessibility
Humanness	0.111	1	0.5
Accessibility	0.143	2	1
Criterion Weight	0.790	0.077	0.133

Table 4 highlights how each category is weighted relative to the others. Performance is deemed significantly more important than both Humanness and Accessibility, with a comparison ratio of 9:1 against Humanness and 7:1 against Accessibility. Conversely, Humanness is considered less important than Accessibility, with a ratio of 0.5:1, indicating that Accessibility is valued twice as much as Humanness. The reciprocal nature of the matrix is evident, as each off-diagonal value corresponds to the reciprocal of its counterpart. For instance, the value comparing Humanness to Performance (0.111) is the reciprocal of the value comparing Performance to Humanness (9). The final row of the table reveals the derived criterion weights, reflecting the overall importance of each category: Performance carries the highest weight at 0.790, indicating its dominant role in the decision-making process. Accessibility follows with a weight of 0.133, while Humanness has the lowest weight at 0.077. These weights suggest that Performance is overwhelmingly prioritized, with Accessibility also considered important, but Humanness is given the least emphasis, showing a strong preference for functional aspects over human-centric qualities.

Conclusion and Recommendations

This study produced a schema-guided dialogue dataset focused on heart disease-related illnesses in Cebuano-Binisaya through user interactions with the Pitik chatbot. The enhancements made to the system resulted in increased accessibility, leading to a notable reduction in performance issues and Gricean Violations. The intrinsic evaluations demonstrated that the improvements allowed for responses to be more effectively tailored to the chatbot’s needs, significantly decreasing Maxim Violations and enhancing accuracy. A conversational scheme was implemented to disambiguate unrecognized responses before proceeding to subsequent questions. Utilizing a reciprocal matrix for pairwise comparisons, as presented in Table 4, the researchers measured the priority of objectives and observed a consistent emphasis on performance. Ultimately, the adoption of the Support Vector Machine (SVM) algorithm in post-intent matching markedly improved the chatbot’s performance and accessibility.

Based on extensive research, several recommendations have been identified. First, the Multinomial Naive Bayes algorithm used for Post-Intent Matching demonstrated inferior performance compared to other algorithms, leading to the recommendation of SVM for text classification, given its superior results with small datasets. During the extrinsic evaluation of the Pitik chatbot, users expressed concerns regarding the lengthy conversational format. Therefore, a button-based scenario is recommended for future iterations to streamline interactions. Additionally, users were dissatisfied with the chatbot’s output, noting a lack of consultations or tips related to heart diseases. Implementing features that allow doctors to edit the risk assessment form addresses this issue, but further recommendations include enabling Pitik to provide medical-related information based on user inputs.

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